

PSEUDOCODE WRITING

← Back

QUESTIONS

Q1 ✓
Student Management System
[BL3 – Apply]
10 marks

Q2 ✓
Library Book Management [BL3 – Apply]
10 marks

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q10 10. Smart Home Device Controller [BL4 – Analyze]

10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
CLASS SmartDevice
METHOD executeCommand(cmd) RETURNS STRING
END
```

```
CLASS SmartLight EXTENDS SmartDevice
state isOn
```

```
CONSTRUCTOR()
isOn <- FALSE
```

[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
CONSTRUCTOR()
    isOn <- FALSE

METHOD executeCommand(cmd) RETURNS STRING
    IF cmd = "on" THEN
        isOn <- TRUE
        RETURN "Light ON"
    ELSE IF cmd = "off" THEN
        isOn <- FALSE
        RETURN "Light OFF"
    ELSE IF cmd = "status" THEN
        IF isOn THEN RETURN "Light is ON"
        ELSE RETURN "Light is OFF"
    ELSE
        RETURN "Invalid Command for Light"
    ENDIF
END

CLASS SmartThermostat EXTENDS SmartDevice
    isOn
    currentTemp
    setPoint

CONSTRUCTOR()
```

[BL3 – Apply]

10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]

10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]

10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]

10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]

10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]

10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]

10 marks

Your Answer

```
CONSTRUCTOR()
```

```
isOn <- FALSE
```

```
currentTemp <- 25
```

```
setPoint <- 25
```

```
METHOD executeCommand(cmd) RETURNS STRING
```

```
IF cmd = "on" THEN
```

```
isOn <- TRUE
```

```
RETURN "Thermostat ON"
```

```
ELSE IF cmd = "off" THEN
```

```
isOn <- FALSE
```

```
RETURN "Thermostat OFF"
```

```
ELSE IF cmd = "status" THEN
```

```
RETURN "Thermostat is " + (isOn ? "ON" : "OFF") + " SetPoint=" + setPoint
```

```
ELSE IF cmd STARTS WITH "set" THEN
```

```
IF isOn = FALSE THEN
```

```
RETURN "Thermostat OFF"
```

```
ENDIF
```

```
newSet <- VALUE AFTER "set"
```

```
setPoint <- newSet
```

```
RETURN "SetPoint Updated"
```

```
ELSE
```

```
RETURN "Invalid Command for Thermostat"
```

```
ENDIF
```

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

END

```
CLASS SmartLock EXTENDS SmartDevice
locked
pin
```

```
CONSTRUCTOR(givenPin)
pin <- givenPin
locked <- TRUE
```

```
METHOD executeCommand(cmd) RETURNS STRING
parts <- SPLIT cmd
action <- LOWERCASE parts[0]
```

```
IF action = "status" THEN
    RETURN "Lock is " + (locked ? "LOCKED" : "UNLOCKED")
ENDIF
```

```
IF LENGTH parts < 2 THEN
    RETURN "PIN required"
ENDIF
```

```
providedPin <- parts[1]
IF providedPin != pin THEN
```

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
IF providedPin != pin THEN
    RETURN "Authentication Failed"
ENDIF

IF action = "lock" THEN
    locked <- TRUE
    RETURN "Locked"
ELSE IF action = "unlock" THEN
    locked <- FALSE
    RETURN "Unlocked"
ELSE
    RETURN "Invalid Command for Lock"
ENDIF
END

CLASS SmartHomeController
    devicesList

    CONSTRUCTOR()
        devicesList <- EMPTY LIST

    METHOD addDevice(device)
        ADD device TO devicesList
```

[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks

Your Answer

```
devicesList <- EMPTY LIST

METHOD addDevice(device)
  ADD device TO devicesList

METHOD controlAll(cmd)
  FOR EACH device IN devicesList DO
    result <- device.executeCommand(cmd)
    PRINT result
  ENDFOR
END

MAIN
  controller <- NEW SmartHomeController
  controller.addDevice(NEW SmartLight)
  controller.addDevice(NEW SmartThermostat)
  controller.addDevice(NEW SmartLock("1234"))

  controller.controlAll("status")
  controller.controlAll("on")
  controller.controlAll("set 26.0")
  controller.controlAll("unlock 1234")
END
```

 Save Answer